

Similar Branching, Different Reach: Dendritic Cable Length Distinguishes between Rat Hippocampal and Neocortical Pyramidal Neurons.

Abstract

The main goal of this study was to compare the dendritic branching patterns of pyramidal neurons in the rat hippocampus and neocortex. This was done using publicly available neuron reconstruction data from NeuroMorpho.org. Twenty neurons from each region were analyzed using the Navis library in Python in Google Colab. Terminal branch (leaf) count, branch count, total dendritic cable length, and a branches-per-length ratio were measured. Neocortical neurons exhibited significantly greater cable length than hippocampal neurons (Welch's t-test, $p = 3.12 \times 10^{-6}$); however, branch count did not differ significantly between the two groups. These findings indicate that pyramidal neurons in the neocortex and hippocampus differ in branching complexity. The hippocampus has more compact, dense branching, whereas the neocortex has long-reaching branches.

Introduction

Pyramidal neurons are excitatory neurons found in several brain regions, including the hippocampus, the cerebral cortex, and the neocortex. Pyramidal neurons play a vital role in brain function, importantly in memory formation in the hippocampus, spatial navigation, and cognition. These neurons are named for their distinct pyramid-shaped cell body, a soma, and two distinct dendritic domains: the basal and the apical dendrites descending from the base and apex of the soma (Spruston, 2008). The extensiveness of these dendrites allows for a larger surface area to receive synaptic input and plays a vital role in determining how the neuron should process and integrate information. All pyramidal neurons share these general characteristics; however, their morphological properties can vary across different cortical regions (Spruston, 2008).

Dendrites are the branching extensions of a neuron that receive signals and carry those signals to the cell body. Dendritic morphology determines the kind of input a neuron receives, thus having a profound impact on information and signal processing (Callaway & Borrell, 2011). Therefore, the morphology of dendrites is particularly important; characteristics such as length, spatial organization, and branching patterns are meaningful for investigating differences in neural dendritic makeup.

The hippocampus and neocortex both contain pyramidal neurons, which makes them useful for investigating differences in their morphology. Despite the similarity in pyramidal structure of neurons in these two regions, they may have different structural specializations that are specific to their location (Spruston, 2008). By comparing the pyramidal neurons in these specific brain regions, we can investigate whether pyramidal neurons with the same overall morphology are organized in the same way throughout the brain regions.

The purpose of the study was to investigate the differences between dendritic patterns in rat hippocampal and neocortical pyramidal neurons using real, publicly available neuron reconstruction data from Neuromorpho.org.

Methods

Data were obtained from the public database, Neuromorpho. Org, using navis Python package. The data were filtered for rat pyramidal neurons in the hippocampus and neocortex, filtering twenty neurons per group. Using 3D reconstruction of each neuron, branch count, leaf count, and total cable length were extracted. Additionally, a branch-per-length ratio was calculated by dividing the number of branches by the cable length of each neuron. This was done to obtain branch density independent of the overall neuron size. Next, group differences were also tested using Welch's independent-samples t-test (`scipy.stats.ttest_ind, equal_var=False`), which does not assume equal variances across groups. Lastly, one group-specific neuron was expressed using `navis.plot2d()` for visual comparison.

Results

Neocortical neurons exhibited significantly more cable length on average (15,209 units) compared to hippocampal neurons (4,496). This difference was found to be statistically significant ($t = -5.74$, $p < 0.001$; Figure 1). However, branch count between the two groups did not differ significantly (98.0 vs. 103.5, $p = 0.83$). Additionally, the branches-per-length ratio also did not exhibit a significant change ($p = 0.081$), although there was an increase in branching density in the hippocampal neurons (**0.0286 vs. 0.0135**), as shown in Table 1.

Table 1. Dendritic branching metrics by brain region

Metric	Hippocampus	Neocortex
Mean branch count	98.00	103.50
Mean leaf count	101.20	112.70
Mean cable length	4495.69	15209.36
Mean branches per length	0.0286	0.0135

Table 1. Dendritic branching metrics by brain region (sample size 20)

The morphology of randomly selected neurons from each brain region (n419, hippocampal; IF1_190301, neocortical; Figure 1) shows the above pattern clearly, with neocortical neurons spanning a larger spatial extent.

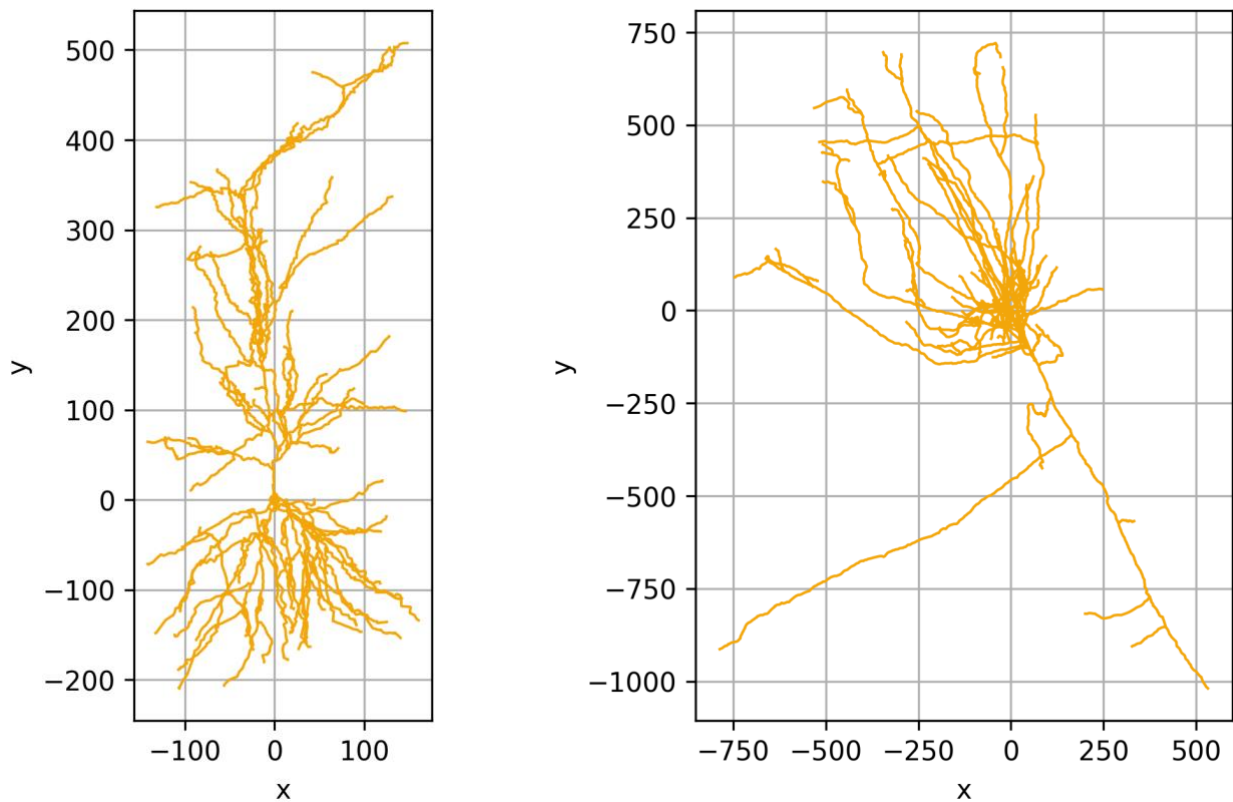


Figure 1. Represents dendritic morphology of one hippocampal neuron (n419, Turner archive, left) and one neocortical neuron (IF1_190301, Staiger archive, right)

However, three hippocampal neurons (n419, n420, n421) had a notably high cable length compared to the rest of the hippocampal neurons; all three came from the same source (Turner) as shown in Figure 2.

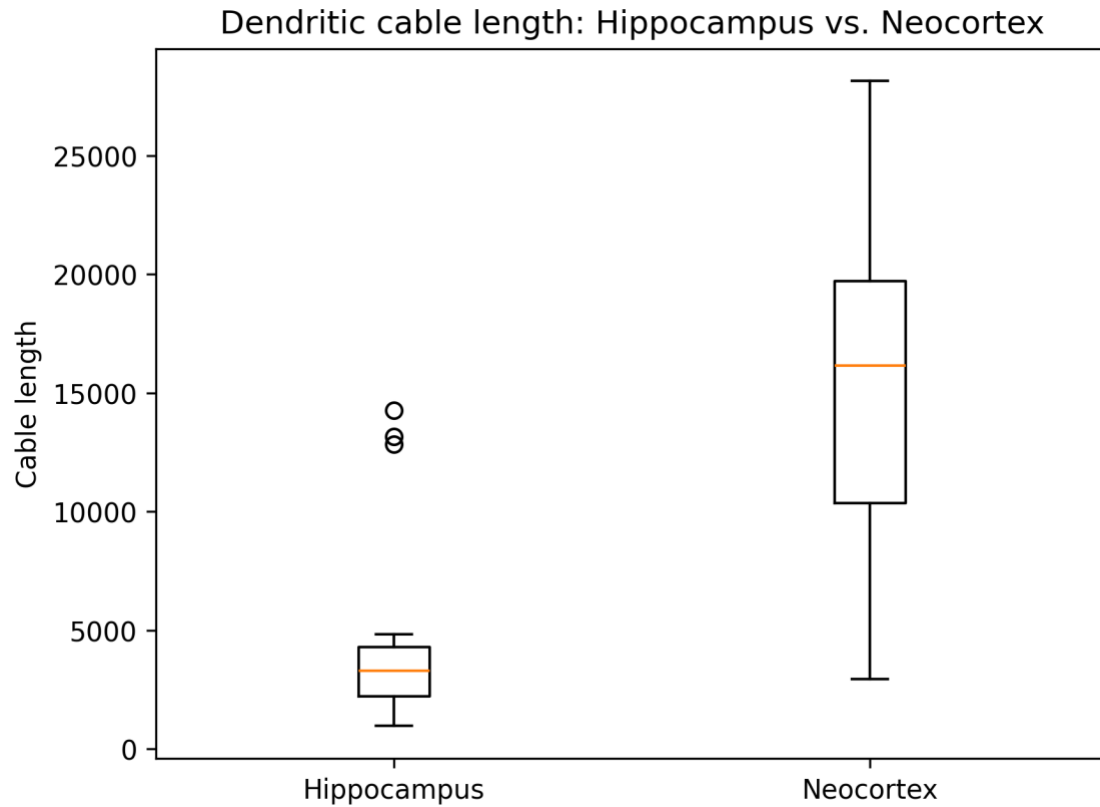


Figure 2. Dendritic cable length comparison between rat hippocampal and neocortical pyramidal neurons (n=20 per group). Boxes show median and interquartile range; circles are the outliers, and whiskers show full range.

Discussion/Limitations

The results indicate that both the hippocampal and the neocortical pyramidal neurons are similar in branching complexity, but they are different in structural dendritic tree-building strategies. The significant difference in cable length and the lack of significant difference in the number of branches indicate that neocortical neurons have a similar number of branches spread over a much larger physical area; in contrast, hippocampal neurons, with a similar number of branches, are packed tightly, spreading over a much shorter physical area. The difference observed here is consistent with the differences between the properties of these two brain regions. Hippocampal neurons such as CA1 are critical for memory encoding and require a tightly packed, constrained

circuit for fast pattern separation. Neocortical neurons send and receive sensory input from a wide range of regions, therefore favoring a long dendritic region.

However, there are several limitations to these conclusions that should be considered. First, the sample size for each group ($n=20$) was relatively small. Second, the data obtained for this project were drawn from multiple independent studies that were performed under different protocols and experimental settings; therefore, it is difficult to conclude whether the differences observed here are due to real biological variation or inter-laboratory differences, such as the three hippocampal outliers with a high cable length (n419, n420, n421) that came from the same archive (Turner). Third, the hippocampal neuronal sample was drawn from a narrow pool of candidates, whereas the neocortical neural sample pool spanned several subregions, such as the barrel region, the secondary motor cortex, and the parietal cortex; thus, it is difficult to conclude or claim that these specific results apply to the “neocortex” and “hippocampus” as unified regions.

Lastly, to predict the differences in pyramidal neurons accurately, larger, more evenly distributed samples must be collected to understand and highlight whether trends such as cable length and branch density observed here are genuine regional differences.

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